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Dynamic Incipient Fault Forecasting for Power Transformers using an LSTM Model

Lin Wang, Tim Littler and Xueqin Liu

***Abstract*—Dissolved gas analysis (DGA) is a traditional approach for power transformer fault diagnostics based on measurement of gas contamination. Hydrocarbon gases generated and dissolved in transformer oil during operation can increase in density as fault conditions predominate. Critical determination of gas concentration changes and assessment trending of dissolved gases for fault prediction and prevention of transformer damage is essential. In this paper, a dynamic fault prediction approach is proposed using a long short-term memory (LSTM) model with intelligent classification to determine the running state of a transformer for prediction and avoidance of potential transformer damage. In the paper, the LSTM model processed DGA data collected from real on-site transformer field measurements and predicts future dissolved gas concentrations in time sequence. Four artificial intelligence diagnostic models (support vector machine (SVM), K-nearest neighbors (KNN), decision tree, and artificial neural network (ANN)) were rendered and used for comparative fault prediction assessment. By comparing experimental results from the different LSTM-based models, this paper asserts that the LSTM-KNN model provides the highest and most reliable prediction accuracy for power transformers.**

***Index Terms*— Dynamic transformer fault prediction, long short-term memory model, intelligence classification methods**.

I. INTRODUCTION

THE transformer represents a significant and high-investment asset which forms a critical component in the power system electricity delivery infrastructure at transmission and distribution levels. The precipitate faults and outages in operational transformers particularly from internal gasification, and insulation changes and breakdown can render high-capital losses and ultimately asset damage. Therefore, condition and health state monitoring for fault tracking and diagnosis are essential in ongoing asset management and maintenance practices [1].

Transformer breakdown can arise from both external and internal damage and faults. Internally, deterioration of insulation includes changes to oil, cellulose paper, and the pressboard structure [2]. Different approaches have been adopted for internal assessment. In [3], a grey correlation diagnosis model was introduced to evaluate the deterioration of internal insulation, based on analysis of pressboard samples using frequency domain spectroscopy. Dissolved gas analysis (DGA) is widely and effectively applied to diagnose and interpret internal transformer condition. Quantities of various gases generated dissolve in the oil along with chemical decomposition of oil and paper insulation. Measurements can be assessed according to gas concentration and ratios, using established methods, including key gas identification, IEC standard cod, Rogers ratio, Duval triangle, and the Dornenburg ratio [4]. These conventional methods rely on human expertise and are limited by specific ratio coding which can cause conflicts and uncertainty in diagnostic outcomes.

In recent years, intelligent models have been considered to eliminate deficiencies in conventional methods and enhance reliability and the accuracy of DGA outcomes and interpretation. Artificial neural network (ANN) [5], support vector machine (SVM) [6], convolutional neural network [7], fuzzy logic [8], adaptive neuro fuzzy inference system [9], k-nearest neighbors (KNN) [10], and gene expression programming [11] have been proposed and applied to transformer fault diagnosis. A semi-supervised auto-encoder based on an auxiliary task was proposed in [12] to determine transformer health degradation and distinguish electrical and thermal faults. The fuzzy association rules combined with a cased-based reasoning method was employed in [13] to assess power transformer fault types. In [14], a feature selection method for DGA data was used with a genetic algorithm (GA), and an improved krill herd (IKH) algorithm was employed to optimize the SVM-based diagnosis model. Bootstrap pre-processing was used in [15] to balance the number of samples in each fault class and genetic programming (GP) extracted the input sample features to improve the classification model. A multinomial logistic regression model was applied to select suitable gases as input to the back-propagation neural networks for fault classification [16]. A probabilistic neural network was optimized by a bat algorithm to enhance the accuracy of transformer fault diagnosis in [17]. To overcome the shortcomings of black box intelligent models, a probability classification model based on a Gaussian process for transformer multi-class faults was proposed in [18]. Furthermore, a deep learning method based on image processing was presented in [19] for detecting transformer winding faults. In [20], a deep belief network (DBN) and stacked denoising auto-encoder were used to select features from transformer vibration signals, combined with relevance vector machines to classify transformer faults.

Most research to date has focused on diagnosing the transformer health condition and assessing a fault type at the instant a fault has occurred. However, prediction of health status is essential for future prevention of potential incipient failure. Jiang et al. applied a Gaussian mixture model to classify transformer health state with

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a hidden Markov model used to describe state changes for time series data [21]. The Gaussian mixture model was used as for unsupervised soft clustering, on the basis that all data were generated from a mixture of Gaussian distributions with unknown parameters. A multi-dimension linear model based on Lasso regression was proposed to observe the change point to predict transformer failure in [22]. The proposed methods in [21] and [22] are useful to carry out preventive maintenance and inspection of power transformers but identification of the types of failures is not achieved. To address this problem, an LSTM model integrated with an improved deep residual shrinkage network, the multi-objective PSO and a random walk strategy was presented in [23]. The proposed model can identify fault signs in advance, but the forecasting accuracy is reduced beyond 2 weeks prediction period. Although the random walk strategy could prevent the PSO trapping in local and global optima, the heuristic algorithm cannot be guaranteed to find the optimal solution and the time required for finding an optimal solution could be significant in some cases. In addition, the fault classification method is based on fault codes in the relevant IEC standard. In comparison to work in [23], this paper has introduced intelligent classifiers to distinguish between normal state and specific fault classes using large-scale DGA field data [18].

Although previous work by the paper authors [18] provides an automated tool for power transformer fault classification using large-scale DGA data by intelligent classifiers, it lacked the predictive capability for future failures. To solve this challenge, in consideration of transformer time sequence information, which is often ignored in common fault diagnostic models and traditional DGA interpretation methods, this paper proposes a dynamic fault diagnosis model to predict future health status for timely assessment of incipient conditions, including determination of the insulation system. The main contributions of this paper are: 1) a running state model is proposed using a time-based long short-term memory (LSTM) model with an intelligent classifier, which focuses on the prediction and classification of health conditions and potential faults for transformer in the future, rather than diagnoses for emerging failures and previous faults investigated in most otherwise published research; 2) real historical and sequential DGA data is used by the deep learning method – LSTM model to help forecast characteristics of dissolved gas concentration. Four typical regression models based on time series forecasting are implemented and discussed in this paper to present a comprehensive comparison to evaluate the reliability and comparative accuracy of the proposed LSTM approach; 3) to realize transformer health condition classification effectively, a large-scale DGA dataset containing 30,000 samples and specific transformer condition (seven fault types and normal state), was employed to develop various computationally intelligent models (SVM, KNN, decision tree, and ANN) to render comparable prediction results based on the predicted gas concentrations. The specific advantages of the proposed approach are: 1) it is able to predict future dissolved gas concentrations by exploiting the powerful deep learning predictive capability; and 2) automatically predict transformer failure types by making use of a large-scale historical DGA database and intelligent classifiers.

This paper is structured as follows. Section II introduces the dissolved gas measurement in transformer oil and considers the LSTM model alongside other regression models. Section III

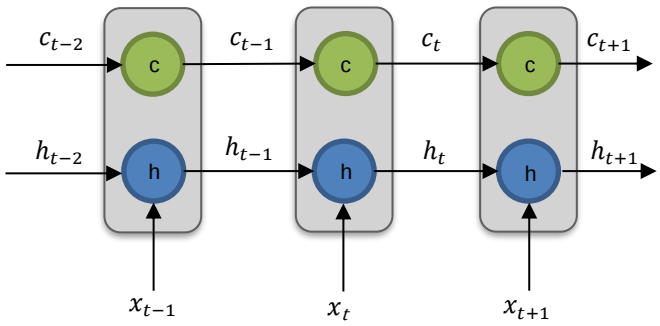


Fig. 1. Sequence structure of LSTM.

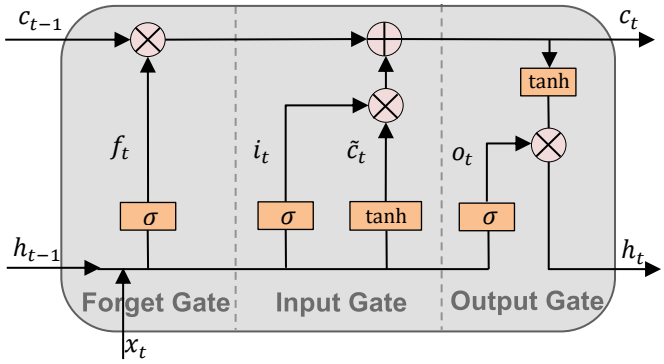


Fig. 2. The structure of LSTM cell.

describes the approach used to develop the LSTM-based model used in this work. Section IV presents and compares the result of the proposed models for one specific transformer. Section V concludes the paper contributions and future work.

II. Time Sequence DGA Data

A. Dissolved Gas Concentrations

The insulation system of an oil-immersed transformer decomposes due to electrical and thermal stresses. Dissolved gases in the transformer insulating oil generally consist of the product of oil decomposition-hydrogen (H_2), methane (CH_4), acetylene (C_2H_2), ethylene (C_2H_4), ethane (C_2H_6) and product of paper decomposition-carbon monoxide (CO), and carbon dioxide (CO_2). Therein, five of these combustible gas concentrations can be exacerbated by thermal and electrical duty and failure: in particular, concentrations of CO and CO_2 are commonly a consequence of aging and deterioration of cellulose paper. The volume of gases and gassing rates can accelerate in response to internal duty, failure and faults, based on overheating, partial discharge, and arcing [24]. Oil chromatographic analysis presents a reliable method for assessing transformer health condition and early detection of incipient issues. Therefore, to limit maintenance investment and resources, forecasting of internal status is essential to prevent incipient failure by using a measure of the change of dissolved gases concentration with respect to historical time sequence DGA data.

B. Long Short-term Memory Model

The LSTM model presents an advanced structure of a recurrent neural network (RNN) to manage gradient vanishing and resolve problems of RNNs and long-term dependency. The LSTM structure is characterized by a sequential model

incorporating historical data [25][26], and is commonly used for speech and text recognition and temperature forecasting. The sequential structure of the LSTM model is presented in Fig. 1. A cell c is introduced and added to control the long term memory compared to RNN, and h presents short term memory. The hidden vector h_{t-1} and memory cell state c_{t-1} from the last time step and the new input x_t at current time step t are taken into consideration to obtain the new output h_t and memory vector c_t.

Fig. 2 describes the cell structure of the LSTM model, divided into three parts: 1) forget gate; 2) input gate; and 3) output gate. The outputs of sigmoidal activation function σ and tanh function are between 0 to 1 and -1 to 1, respectively. The circle points are vector multiplication and addition operations. The forget gate determines the amount of previous information remained to influence the current cell state c_t. f_t is the gate state wherein 0 and 1 represent to forget and keep the information, respectively. The input gate is used to add and update new information to the cell, i_t is the input gate state. The sigmoidal layer is to select the useful information from a new input for the update. The tanh layer adds or removes new information to the cell state at the current time step. The new cell state c_t can be obtained based on the outputs (i_t, $\tilde{c}_t$) of the two neural network layers. The output gate is applied to determine the information selected from the cell for output, and o_t is the output gate state and h_t is the output of new input x_t. The corresponding calculation can be derived from (1) to (6).

$$f_t = \sigma\big(W_f \cdot [h_{t-1}, x_t] + b_f\big) \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{2}$$

$$\tilde{c}_t = tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{3}$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t \tag{4}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_0) \tag{5}$$

$$h_t = o_t * tanh(c_t) \tag{6}$$

where W_f, W_i, W_c and W_o represent the weight matrices, and b_f, b_i, b_c and b_o which are biases of the forget gate, input gate and output gate.

C. Regression Models

Four classical regression models are used for comparative assessment of the LSTM model to predict the dissolved gas concentration according to time series' DGA datasets. Regression models are: Support Vector Regression (SVR), the regression tree, Gaussian Process Regression (GPR), and regression neural network.

1) Support Vector Regression

A Support Vector Machine (SVM) is widely employed for classification problems, which is called support vector regression if used for regression. SVR applies a kernel function to map the input feature space of training samples into higher dimensions [27]. The hyperplane defined in the SVM for separating classes is used to predict the continuous real numbers in SVR. A non-linear SVR is implemented in this paper that employs an ε-insensive loss function. The dual formulation is given by:

$$L(\alpha) = \max_{\alpha,\alpha^*} \sum_{i=1}^{N} y_i(\alpha_i - \alpha_i^*) - \varepsilon \sum_{i=1}^{N} (\alpha_i + \alpha_i^*) - \frac{1}{2} \sum_{i,j=1}^{N} (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) \tag{7}$$

Subject to:

$$\sum_{i=1}^{N} (\alpha_i - \alpha_i^*) = 0, \quad 0 \le \alpha_i, \alpha_i^* \le C$$

where α_i and α_i^* are Lagrange multipliers, and ε represents the margin threshold. C is a pre-defined value refers to the box constraint. The prediction function for new value can be obtained:

$$f(x) = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) K(x_i, x) + b \tag{8}$$

2) Regression Tree

The regression tree is to realize regression with the tree module. It also consists of root, node and leaf identical to a classification tree. The data from a root node moves down to next node along with the branch based on feature selection, until it reaches the leaf. Each leaf node outputs a predictive value using the regression function in it, which is the average value of the training samples including in this leaf [28]. The classification and regression tree (CART) is used in this paper. For regression problems, the feature selection and criteria of a split are determined by the mean square error or mean absolute error rather than the Gini index that is employed for classification.

3) Gaussian Process Regression

GPR is recognized as a nonparametric model with a Bayesian framework. A Gaussian process prior is defined to calculate the posterior of the data. The probability distribution of all possible functions used to fit the observed data is inferred, and the prediction for new data is derived by the functions [29]. To smooth the functions, a covariance function is introduced to express prior knowledge. A *Matern* kernel function is adopted in this paper and is described as below:

$$K(x_i, x_j) = \sigma_f^2 (1 + \frac{\sqrt{5}r}{\sigma_l} + \frac{5r^2}{3\sigma_l^2}) exp(-\frac{\sqrt{5}r}{\sigma_l}) \tag{9}$$

where r is the Euclidean distance between x_i and x_j, expressed by $r = \sqrt{(x_i - x_j)^T (x_i - x_j)}$. Also, σ_f and σ_l represent the characteristic length scale and the signal standard deviation, respectively.

4) Neural Network

A neural network (NN) is usually applied for classification. The classical NN model consists of three fully connected layers: 1) input layer; 2) hidden layer; and 3) output layer [30]. An activation function is rendered for each layer to return a character vector. The output of the last layer is used as the input to the next layer, which is used to multiply a weight matrix and adds a bias in the layer. The regression model can be realized by replacing the activation function of the output layer by a

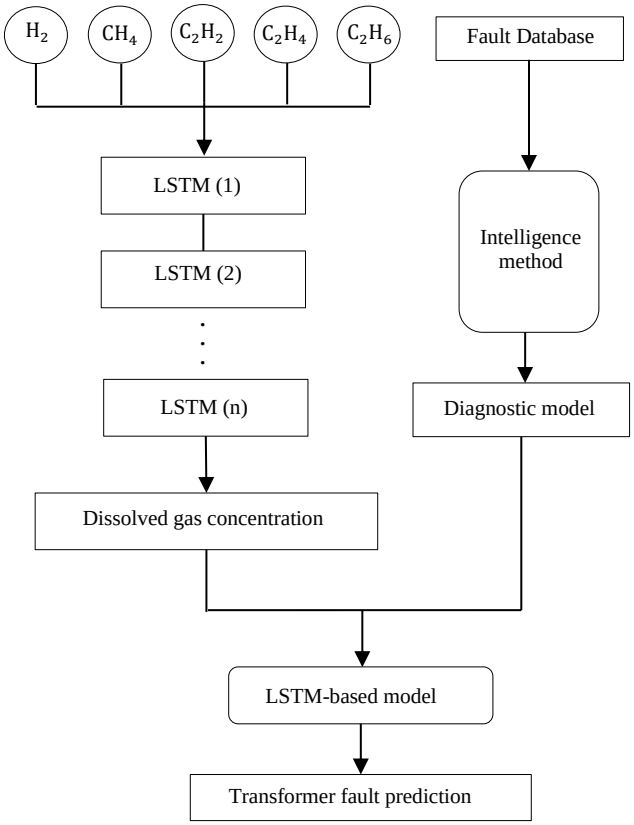


Fig. 3. The flowchart of dynamic fault prediction model.

linear activation function, to map the observed data to a continuous value.

III. Dynamic Fault Prediction Model

Due to a correlation of data from transformer dissolved gases and fault types, it is reliable and important to monitor the change of gas concentrations along with time index for transformer health state assessment. Therefore, the historical DGA information is used to develop a dynamic forecasting model for dissolved gas concentration based on the LSTM model. The intelligent models are implemented with adequate DGA datasets labelled with specific fault types, and used with predicted gas concentrations to forecast and diagnose potential transformer failure and fault emergence. Four artificial intelligence (AI) models are used for comparative assessment based on the data sets from five combustible dissolved, to validate the predictive accuracy of proposed LSTM-based model. The flowchart of the LSTM-based predictive model is described in Fig. 3. Resolution steps are as follows:

1) Collect historical information along with time indices from five combustible dissolved gases concentration (H_2, CH_4, C_2H_2, C_2H_4, C_2H_6).

2) Develop the LSTM model with *n* time series DGA datasets. Back propagation with a time index is used for feature selection, error calculation and the gradient of each weight. A forward pass algorithm is employed to calculate the loss function. The output is a forecast trend of the five gas concentrations.

3) Four AI models (SVM, KNN, decision tree, and ANN) are trained with the DGA datasets extracted from a constructed fault database [31]. Eight transformer health conditions can be detected by the model, which are: low-energy discharge (D1); high-energy discharge (D2); a combination fault of discharge fault and thermal fault; partial discharge (PD); low temperature fault (T1); medium temperature fault (T2); high temperature fault (T3); and normal state (NS). Five gas concentration data sets are used as the input, and transformer health state is generated as the output.

4) To determine potential future failure and ascertain risks, a forecast of the five key gas concentrations using step 2) above is obtained, based on previously obtained DGA transformer data, and used to train an intelligent classification model for condition assessment.

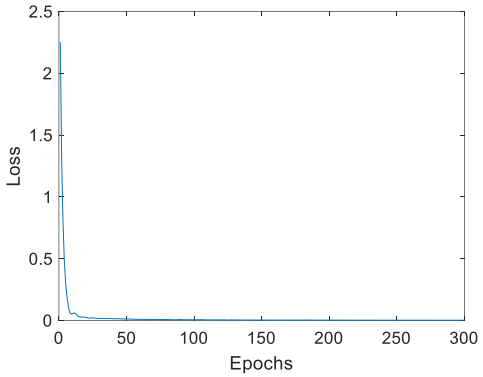


Fig. 4. LSTM model training progress.

TABLE I
RMSE AND LOSS WITH DIFFERENT NUMBER OF NEURONS IN LSTM

Number of neurons	RMSE	Loss
2	0.1275	0.0081
4	0.1025	0.0053
6	0.0823	0.0034
8	0.0794	0.0032
10	0.0678	0.0023
12	0.0793	0.0031

TABLE II
THE MAXIMUM RELATIVE AND AVERAGE RELATIVE ERRORS OF EACH GAS CONCENTRATION

	H_2	CH_4	C_2H_2	C_2H_4	C_2H_6
MRE	0.05	0.13	0.12	0.24	0.12
ARE	0.01	0.03	0.03	0.03	0.03

IV. Result and Discussion

A. Gas Concentration Prediction

In this paper, DGA data was collected from a distribution level, oil-immersed transformer with 4 hours sampling interval. 1764 sample points from February 27, 2014 to January 16, 2015 were used to train the LSTM model, and the gas concentration trend in the next seven weeks from January 17, 2015 to March 6, 2015 was predicted with the trained LSTM. The root mean squared error (RMSE) and loss function are introduced to assess the performance of the LSTM model. The initial learning rate was 0.02, the batch size was 8, and the number of iterations set to 300. The *Adam* optimization algorithm was employed to

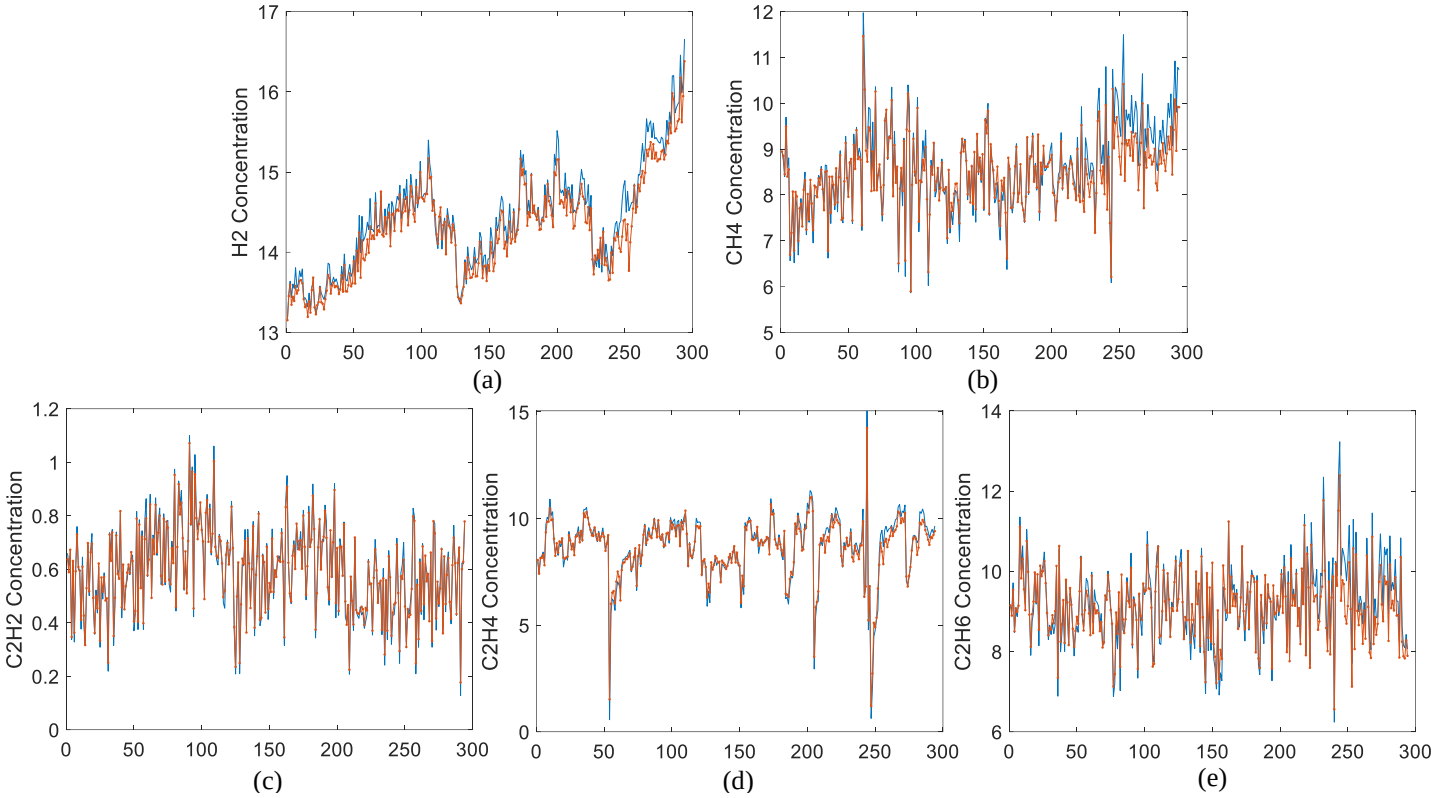


Fig. 5. Comparison of predicted dissolved gas concentrations (red line) and observed values (blue line). (a) H_2 concentration. (b) CH_4 concentration. (c) C_2H_2 concentration. (d) C_2H_4 concentration. (e) C_2H_6 concentration.

TABLE III
MAE, RMSE AND MSE OF FIVE PREDICTIVE MODELS

Model		H_2	CH_4	C_2H_2	C_2H_4	C_2H_6
LSTM	MAE	**0.14**	**0.24**	**0.02**	**0.21**	**0.25**
	RMSE	**0.17**	**0.33**	**0.03**	**0.28**	**0.33**
	MSE	**0.03**	**0.11**	**6.3e-4**	**0.08**	**0.11**
SVR	MAE	0.54	0.76	0.07	0.95	0.41
	RMSE	0.69	1.01	0.09	1.15	0.64
	MSE	0.48	1.02	0.01	1.34	0.51
Decision tree	MAE	0.32	0.52	0.10	1.22	0.51
	RMSE	0.61	0.75	0.13	1.58	0.68
	MSE	0.37	0.56	0.02	2.49	0.46
GPR	MAE	0.78	0.57	0.07	1.46	0.75
	RMSE	1.09	0.82	0.09	2.06	1.04
	MSE	1.18	0.68	0.01	4.25	1.09
BNN	MAE	0.97	0.53	0.09	1.15	0.64
	RMSE	1.22	0.75	0.12	1.53	0.90
	MSE	1.48	0.56	0.01	2.35	0.81

determine the stochastic gradient descent. To avoid and eliminate the disturbance of noise influence the performance of LSTM, the DGA data samples were pre-proceed for denoising based on Finite Impulse Response (FIR) filter. Table I presents the RMSE and loss, which vary with the number of neurons: it is apparent that the LSTM model with 10 neurons rendered the lowest RMSE and loss, 0.0678 and 0.0023, respectively. The convergence curve of the LSTM model with 10 neurons is shown in Fig. 4, which plots loss as a function of the number of epochs. It is clear that the loss tends to stabilise and obtain as minimum value of 0.002 after 282 epochs have elapsed.

The prediction of five dissolved gas concentrations (H_2, CH_4, C_2H_2, C_2H_4, and C_2H_6) in comparison with observed values determined for a projection of seven weeks ahead is presented in Fig. 5. It is apparent that the five gases correlate well with the original data and have followed the target trends. Relative error is calculated to describe the uncertainty of gas concentration prediction in (10). Table II presents the maximum relative errors (MRE) and the average relative errors (ARE) for each gas concentration. The relative error of H_2 and C_2H_4 prediction is presented in Fig. 6. The fluctuation of H_2 trend is more stable without outliers in comparison with C_2H_4.

$$Relative\ Error\ = \left|\frac{\tilde{x}_t - x_t}{x_t}\right| \times 100\% \tag{10}$$

where $\tilde{x}_t$ and x_t represent predicted and actual value, respectively.

$$MAE\ = \frac{1}{N}\sum_{t=1}^{N}|\tilde{x}_t - x_t| \tag{11}$$

$$RMSE\ = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(\tilde{x}_t - x_t)^2} \tag{12}$$

$$MSE = \frac{1}{N}\sum_{t=1}^{N}(\tilde{x}_t - x_t)^2 \tag{13}$$

To analyse and evaluate the performance of the LSTM model, three indices, the mean absolute error (MAE), RMSE, and mean squared error (MSE) are introduced in this paper. The calculation formulas are described in (11)-(13). The SVR,

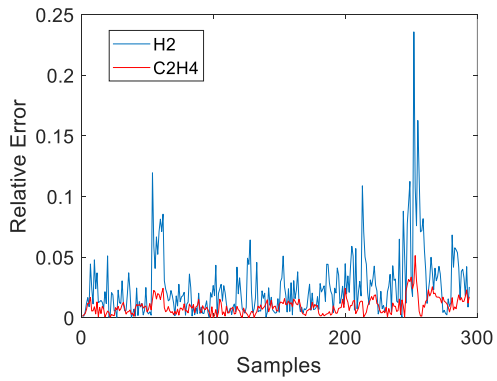


Fig. 6. The relative error of predicted H_2 and C_2H_4 gas concentration.

Regression Tree, GPR and Bilayered Neural Network (BNN) were all compared with the proposed LSTM model. SVR employed a Gaussian kernel function with the kernel scale equal to 0.5. The Regression Tree, referred to as a decision tree, had the minimum leaf set to 3. A *Matern* 5/2 kernel function was used in GPR. The number of fully connected layers was 2 in the BNN, where the first layer size and second layer size were both 8 using the *ReLU* activation function. Table III tabulates the evaluation indices of five dissolved gases with each of the different models. The LSTM model performed better than the four comparative models in terms of MAE, RMSE and MSE. The smallest MAE and RMSE resulted from H_2 concentration determination with 0.14 and 0.17, respectively attained for the LSTM model, followed by the regression tree model with MAE of 0.32 and RMSE of 0.61. The BNN model rendered the largest MAE and MSE, with 0.97 and 1.48, respectively attained. The prediction result for gas CH_4 had an MAE of 0.24, and MSE of 0.11, from evaluation errors in the LSTM model, the regression tree, GPR and BNN outcomes were similar. For prediction of the C_2H_2 concentration, LSTM presented a small error, with MAE of 0.02, RMSE of 0.03, and MSE of < 0.01. SVR and regression tree rendered similar results with GPR and BNN, respectively. The MSE for gas concentration C_2H_4 was 0.08 with the LSTM model, 1.26 and 4.17 lower than SVR and GPR, respectively. The RMSE of the C_2H_6 prediction was 0.33 in the LSTM model, with > 0.60 in the comparative models. The LSTM model was capable of predicting dissolved gas concentrations for several weeks ahead, better than the four regression models in comparison, using the historical DGA data information.

B. Performance of Classification Models

A total of 30,000 transformer DGA datasets, including sets with seven different fault types and normal states, were used to train the intelligent diagnostic models: SVM, KNN, decision tree, and ANN. The training data consisted of: 1914 cases with discharge faults; 22,783 cases with thermal faults; 303 cases with combined discharge and thermal faults; and 5000 healthy cases. The number of datasets with thermal faults greatly exceeded the other fault cases as these are more frequent occurrences. Tables IV-VII tabulate the confusion matrix of the four classification models using the 30,000 training datasets. The value on the diagonal represents the number of datasets in which the fault type was diagnosed exactly. The decision tree rendered the best performance with 97.8% classification accuracy, 3.3% accuracy greater than ANN, followed SVM

TABLE IV
THE SVM CONFUSION MATRIX

Predicted class	True class							
	D1	D2	DT	PD	T1	T2	T3	NS
D1	526	19	0	0	0	0	0	0
D2	35	1275	0	0	0	0	0	0
DT	0	1	242	0	1	8	0	8
PD	0	0	0	14	0	0	0	0
T1	0	0	2	20	2819	15	0	21
T2	0	0	8	0	26	5933	91	89
T3	1	2	27	0	1	68	13691	108
NS	2	3	24	16	39	31	60	4774
Accuracy	97.6%							

TABLE V
THE KNN CONFUSION MATRIX

Predicted class	True class							
	D1	D2	DT	PD	T1	T2	T3	NS
D1	444	83	0	0	0	0	0	0
D2	120	1217	0	0	0	0	0	0
DT	0	0	254	0	0	10	2	33
PD	0	0	0	14	10	0	0	4
T1	0	0	3	22	2763	66	0	32
T2	0	0	18	2	83	5886	44	91
T3	0	0	4	0	0	34	13736	74
NS	0	0	24	12	30	59	60	4766
Accuracy	96.9%							

TABLE VI
THE DECISION TREE CONFUSION MATRIX

Predicted class	True class							
	D1	D2	DT	PD	T1	T2	T3	NS
D1	505	31	0	0	0	0	0	0
D2	59	1266	0	0	0	2	0	0
DT	0	2	258	0	3	20	2	10
PD	0	0	0	28	11	1	0	2
T1	0	0	6	17	2775	71	0	35
T2	0	1	18	2	83	5869	51	90
T3	0	0	13	1	0	50	13780	5
NS	0	0	8	2	14	42	9	4858
Accuracy	97.8%							

TABLE VII
THE ANN CONFUSION MATRIX

Predicted class	True class							
	D1	D2	DT	PD	T1	T2	T3	NS
D1	464	77	0	0	0	0	0	0
D2	58	1219	21	0	0	0	0	0
DT	0	0	0	0	0	0	0	0
PD	0	0	0	0	0	0	0	0
T1	42	0	4	36	2521	11	0	75
T2	0	0	16	0	316	5724	98	70
T3	0	0	1	0	0	48	13650	90
NS	0	4	261	14	49	272	94	4765
Accuracy	94.5%							

TABLE VIII
CLASSIFICATION ACCURACY FOR DIFFERENT FAULT TYPES (%)

Classifier	Discharge faults	Thermal faults	Combined fault	Healthy state
SVM	94.8	98.5	79.9	95.5
KNN	87.5	98.3	83.8	95.3
Decision tree	94.0	98.4	85.1	97.2
ANN	87.9	96.1	0	95.3

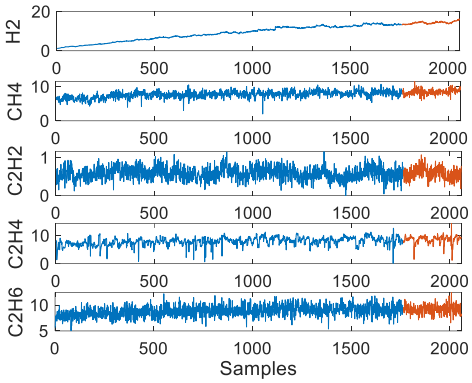


Fig. 7. The trends of dissolved gas concentrations: the blue lines represent the training data samples; the red lines represent the gas concentration predictions for seven weeks ahead.

TABLE IX
FAULT PREDICTION ACCURACY OF THE CLASSIFICATION MODELS

Prediction Period (week)	Accuracy (%)			
	SVM	KNN	Decision tree	ANN
1st	76.2	83.3	92.9	83.3
2nd	95.2	88.1	95.2	95.2
3rd	64.3	81.0	78.6	85.7
4th	83.3	90.5	92.9	90.5
5th	85.7	95.2	97.6	95.2
6th	78.6	83.3	81.0	83.3
7th	81.0	78.6	71.4	69.0
Total average	80.6	85.7	87.1	86.1

with an accuracy of 97.6%. Classification accuracy performance for each model with the various fault types was similar, apart from ANN: the ANN model classified all the datasets with DT and PD fault incorrectly.

Table VIII tabulates the diagnosis accuracy for discharge faults (D1, D2, and PD), thermal faults (T1, T2, and T3), the combined fault (DT), and the healthy state of transformer using the four classification models. The best accuracy for discharge and thermal faults was attained by the SVM model, with 94.8% and 98.5%, respectively. The ANN model rendered a lower accuracy for thermal faults with 96.1% attainment. The KNN model had an 87.5% accuracy for discharge faults, a 7.3% accuracy difference compared to the SVM model. Meanwhile, the decision tree classified the combination fault and transformer healthy state with the highest accuracy of 85.1% and 97.2%, respectively.

C. Transformer Condition Prediction

The change trend of dissolved gas concentrations including the original training DGA datasets and predicted concentrations for a future seven weeks is presented in Fig. 7. The five dissolved gas concentrations of 1764 historical DGA data samples used in this work are represented by the blue lines. It is observed that all gas concentrations excluding C_2H_2 present an upward trend overall, particularly for H_2 while a relatively stable change emerges with concentration of C_2H_2: this appears since the volume of C_2H_2 usually increases significantly with higher temperatures (>700℃), especially >800℃. In general, the prediction results for all five dissolved gases shown by the red lines fit the overall trends and the fluctuations of the training samples well.

TABLE X
FAULT CASES FOR EACH FUTURE WEEK PREDICTION

1st week	SVM	KNN	Decision tree	ANN	True state
DT	0	0	1	0	0
T2	0	0	0	1	0
T3	10	13	4	10	6
NS	32	29	37	31	36
Actual fault rate			14.3%		
2nd week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	3	0	0	1
T2	2	3	0	1	0
T3	1	2	3	0	2
NS	39	34	39	41	39
Actual fault rate			7.1%		
3rd week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	0	4	0	0
T2	1	1	1	1	1
T3	16	12	0	6	4
NS	25	29	37	35	37
Actual fault rate			11.9%		
4th week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	1	3	0	1
T2	0	0	0	0	0
T3	5	3	0	4	1
NS	37	38	39	38	40
Actual fault rate			4.8%		
5th week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	0	1	0	0
T2	0	1	0	0	0
T3	15	8	9	9	9
NS	27	33	32	33	33
Actual fault rate			21.4%		
6th week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	2	0	0	3
T2	4	6	1	3	5
T3	8	7	2	5	3
NS	30	27	39	34	31
Actual fault rate			26.2%		
7th week	**SVM**	**KNN**	**Decision tree**	**ANN**	**True state**
DT	0	0	1	0	0
T2	4	4	3	6	11
T3	10	8	2	2	5
NS	28	30	36	34	26
Actual fault rate			38.1%		

To form a dynamic transformer condition prediction model, the evaluated gas concentrations are input to the four classification models developed in Section IV-B. The predicted fault type is compared with the observed value for future seven weeks forecasts: Table IX presents prediction accuracies for the future seven weeks. Most of the diagnostic accuracies for the first five (of seven) weeks using the decision tree and the ANN models are > 90%, excluding a 78.6% accuracy (for the third future week) using the decision tree, while the ANN presented accuracies of 83.3% and 85.7% for the first and third weeks. However, the classification accuracies for the future sixth week

TABLE XI
CONFUSION MATRIX OF THE TRUE AND PREDICTED CLASS FOR THE FUTURE SEVEN WEEKS USING VARIOUS CLASSIFIERS

Classifier	Predicted class	True class			
		DT	T2	T3	NS
SVM	DT	**0**	0	0	0
	T2	0	**10**	0	1
	T3	0	5	**23**	37
	NS	5	4	5	**204**
KNN	DT	**4**	0	0	2
	T2	1	**12**	0	2
	T3	0	1	**25**	27
	NS	0	6	3	**211**
Decision tree	DT	**1**	0	1	8
	T2	0	**6**	0	2
	T3	0	0	**17**	0
	NS	4	13	10	**232**
ANN	DT	**0**	0	0	0
	T2	0	**8**	1	30
	T3	0	0	**21**	15
	NS	5	11	6	**224**

and seventh week decreased to < 85% and < 72%, respectively. The decision tree had the best performance in the first future week (92.9% accuracy), followed by the ANN and the KNN with 83.3% accuracy. Models other than KNN had 95.2% accuracy in the second predicted week. An 81.0% accuracy was attained by KNN in the third week, 4.7% lower than that of the ANN. For the fourth and fifth future week predictions, the highest prediction accuracy was obtained with the decision tree model, slightly higher than that of the KNN and ANN. All classification models had similar accuracy for the sixth week. The lowest accuracy for the seventh week was shown by ANN with 69.0%. The overall accuracy of SVM, KNN, the decision tree, and the ANN models was 80.6%, 85.7%, 87.1% and 86.1%, respectively. However, the KNN model rendered a more stable prediction accuracy of $86.9 \pm 8.3\%$ during the seven-week prediction period, in comparison with other models which had significant differences between the highest and the lowest accuracy.

The fault rate for the seven future weeks and true health state are tabulated in Table X. To clarify, actual faults were assessed by a hybrid method combining the Roger's ratio and Duval triangle method to construct a fault database, which was proposed in our previous work [18][31]. In most cases, the transformer condition was healthy; only rare cases with DT faults and some cases with thermal faults, particularly for T3 occurred, which matched the actual fault type distribution. The fourth and seventh weeks have the lowest and highest actual fault rates, 4.8% and 38.1%, respectively. The decision tree model predicted the fault rate in the first four future weeks more accurately than comparative models, which obtained the same fault rate as the actual values (11.9% and 7.1%) in the second and third weeks. In the fifth future week, the probability of fault occurrence was predicted as 21.4% using the KNN and ANN models, consistent with the actual fault rate, followed by the decision tree model (23.8%). The fault probability evaluated by the SVM model was close to the actual value in the sixth and seventh future week predictions, with 28.5% and 33.3%, respectively, followed by the KNN model.

Table XI presents the specific distribution of fault classification for the future seven-week period. All five samples with the DT fault were predicted incorrectly as a normal state by the SVM and the ANN models, while four samples therein were detected accurately with the KNN. Meanwhile, 12 out of 18 cases with the T2 fault, and 25 out of 28 cases with the T3 fault were classified correctly based on the KNN model. The KNN model had a better performance on fault diagnosis than the others, followed by SVM. The decision tree and ANN models performed better on classification of the normal state. The number of healthy samples (242 cases) was greater than the number of fault samples (52 cases), resulting in a higher overall accuracy rendered by the decision tree and ANN models in some instances shown in Table IX. However, prediction of potential faults was most reliably delivered by the KNN model.

V. CONCLUSION

Accurate and preemptive determination of the operational state and condition of a power transformer is critical to limit and avoid failure. Moreover, asset management practices require precise information to develop effective maintenance policies and strategies. In response to the excellent performance and high accuracy of deep learning methods presented and published in various areas over recent years, a dynamic prediction model for latent and potential future faults is proposed in this paper based on an LSTM and intelligent classifier. The main conclusions from this paper are as follows.

1) A time index and historical dissolved gas concentrations contained in 1764 real DGA data samples over 11 months have been adopted in the proposed LSTM model to forecast future gas concentration trending. Several regression models – the SVR, regression tree, GPR, and BNN were employed to provide a comprehensive comparison with the LSTM model. Three performance indices – MAE, RMSE, and MSE were used for result analyses. The LSTM model exhibited a better performance than the comparator models, with an MAE of < 0.26, RMSE of < 0.34, and MSE of < 0.12. Meanwhile, a good fitness was presented between the predicted concentration trend and target value, which validates the LSTM model as robustly effective in predicting future dissolved gas concentration accurately.

2) To diagnose specific transformer health conditions including the seven different fault types and normal state used in this paper, four intelligent computational models were trained with a large-scale DGA dataset, which contributed to a significant improvement in diagnostic accuracy. The SVM, KNN, and decision tree models had a high classification accuracy (> 96.8%) for training data, while the ANN model delivered 94.5% accuracy.

3) A comparison of performance for forecasting and classifying future transformer health conditions over the seven week period adopted in this paper and using different diagnostic models has been presented. Although the KNN, decision tree and ANN models performed with comparable outcomes for (overall) prediction accuracy in the range 85.6% to 87.2%, the accuracy of the decision tree and ANN models was mainly as a result of correct classification of the majority class (data samples with the normal state). The main contribution of this

paper is in proposing a method by which it is possible to forecast future faults effectively, and as such, classification accuracy of fault samples is important. The experimental results demonstrate that the KNN model predicted fault conditions in most instances correctly, particularly for the rare DT fault, and delivered a better performance in comparison to other relevant classifiers. Accordingly, the LSTM-KNN model provides reliable and accurate prediction for determination of future and potential faults in power transformers.

4) The overall accuracy of the classification model is unreliable if there is a great difference in the number of samples between classes. Hence, acknowledging the diagnosis accuracy of each class for result analyses is necessary and significant.

The transformer investigated in this paper contained mainly thermal faults, though no discharge faults. Future work will therefore focus on assessment of a broader range of transformer and fault types to evaluate the general applicability of the proposed LSTM-KNN model. Furthermore, training samples for the classification model used in this paper had an imbalanced distribution among the fault classes. To further improve prediction accuracy, solutions for the imbalanced training data will be therefore investigated in the future.

REFERENCES

[1] S. Li, G. Wu, B. Gao, C. Hao, D. Xin and X. Yin, "Interpretation of DGA for transformer fault diagnosis with complementary SaE- Elm and arctangent transform," *IEEE Trans. Dielectr. Electr. Insul*., vol. 23, no. 1, pp. 586-595, Feb. 2016.

[2] S. J. Lee, Y. M. Kim, H. D. Seo, J. R. Jung, H. J. Yang and M. Duval, "New methods of DGA diagnosis using IEC TC 10 and related databases Part 2: application of relative content of fault gases," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 20, no. 2, pp. 691-696, Apr. 2013.

[3] J. Gao, L. Yang, Y. Wang, X. Liu, Y. Lv and H. Zheng, "Condition diagnosis of transformer oil-paper insulation using dielectric response fingerprint characteristics," *IEEE Trans. Dielectr. Electr. Insul*., vol. 23, no. 2, pp. 1207-1218, Apr. 2016.

[4] *IEEE Guide for the Interpretation of Gases Generated in Oil-Immersed Transformers*, IEEE Std C57.104-2008, 2009.

[5] S. S. M. Ghoneim, I. B. M. Taha and N. I. Elkalashy, "Integrated ANN-based proactive fault diagnostic scheme for power transformers using dissolved gas analysis," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 23, no. 3, pp. 1838-1845, Jun. 2016.

[6] J. Li, Q. Zhang, K. Wang, J. Wang, T. Zhou and Y. Zhang, "Optimal dissolved gas ratios selected by genetic algorithm for power transformer fault diagnosis based on support vector machine," *IEEE Trans. Dielectr. Electr. Insul*, vol. 23, no. 2, pp. 1198-1206, Apr. 2016.

[7] I. B. M. Taha, S. Ibrahim and D. -E. A. Mansour, "Power transformer fault diagnosis based on DGA using a convolutional neural network with noise in measurements," *IEEE Access*, vol. 9, pp. 111162-111170, Aug. 2021.

[8] M. Noori, R. Effatnejad and P. Hajihosseini, "Using dissolved gas analysis results to detect and isolate the internal faults of power transformers by applying a fuzzy logic method," *IET Gener. Transmiss. Distrib.*, vol. 11, no. 10, pp. 2721-2729, Jul. 2017.

[9] S.A. Khan, Md.D. Equbal and T. Islam, "A comprehensive comparative study of DGA based transformer fault diagnosis using fuzzy logic and ANFIS models," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 22, no. 1, pp. 590-596, Feb. 2015.

[10] O. Kherif, Y. Benmahamed, M. Teguar, A. Boubakeur and S. S. M. Ghoneim, "Accuracy improvement of power transformer faults diagnostic using KNN classifier with decision tree principle," *IEEE Access*, vol. 9, pp. 81693-81701, Jun. 2021.

[11] H. Malik and S. Mishra, "Application of gene expression programming (GEP) in power transformers fault diagnosis using DGA," *IEEE Trans. Ind. Appl.*, vol. 52, no. 6, pp. 4556-4565, Aug. 2016.

[12] S. Kim *et al*., "A semi-supervised autoencoder with an auxiliary task s(SAAT) for power transformer fault diagnosis using dissolved gas analysis," *IEEE Access*, vol. 8, pp. 178295-178310, Sep. 2020.

[13] C. Guo *et al*., "Transformer failure diagnosis using fuzzy association rule mining combined with case-based reasoning," *IET Gener. Transm. Distrib*., vol. 14, no. 11, pp. 2202–2208, June 2020.

[14] Y. Y. Zhang *et al*., "A fault diagnosis model of power transformers based on dissolved gas analysis features selection and improved krill herd algorithm optimized support vector machine," *IEEE Access*, 2019.

[15] A. Shintemirov, W. Tang and Q. H. Wu, "Power transformer fault classification based on dissolved gas analysis by implementing bootstrap and genetic programming," *IEEE Trans. Syst. Man Cybern. C Appl. Rev.*, vol. 39, no. 1, pp. 69-79, Jan. 2009.

[16] M. Yang and L. Hu, "Intelligent fault types diagnostic system for dissolved gas analysis of oil-immersed power transformer," *IEEE Trans. Dielectr. Electr. Insul*., vol. 20, no. 6, pp. 2317-2324, Dec. 2013.

[17] X. Yang, W. Chen, A. Li, C. Yang, Z. Xie and H. Dong, "BA-PNN-based methods for power transformer fault diagnosis," *Adv. Eng. Inform*., vol. 39, pp. 178-185, Jan. 2019.

[18] L. Wang, T. Littler and X. Liu, "Gaussian process multi-class classification for transformer fault diagnosis using dissolved gas analysis," *IEEE Trans. Dielectr. Electr. Insul*., vol. 28, no. 5, pp. 1703-1712, Oct. 2021.

[19] A. Moradzadeh, H. Moayyed, B. Mohammadi-Ivatloo, G. B. Gharehpetian and A. P. Aguiar, "Turn-to-turn short circuit fault localization in transformer winding via image processing and deep learning method," *IEEE Trans. Industr. Inform*., vol. 18, no. 7, pp. 4417-4426, Jul. 2022.

[20] C. Zhang, Y. He, B. Du, L. Yuan, B. Li and S. Jiang, "Transformer fault diagnosis method using IoT based monitoring system and ensemble machine learning," *Future Gener. Comput. Syst.*, vol. 108, pp. 533-545, Mar. 2020.

[21] J. Jiang, R. Chen, M. Chen, W. Wang and C. Zhang, "Dynamic fault prediction of power transformers based on hidden Markov model of dissolved gases analysis," *IEEE Trans. Power Del.*, vol. 34, no. 4, pp. 1393-1400, Aug. 2019.

[22] J. Jiang, R. Chen, C. Zhang, M. Chen, X. Li and G. Ma, "Dynamic fault prediction of power transformers based on Lasso regression and change point detection by dissolved gas analysis," *IEEE Trans. Dielectr. Elect. Insul*., vol. 27, pp. 2130-2137, Dec. 2020.

[23] X. Ma, H. Hu and Y. Shang, "A new method for transformer fault prediction based on multifeature enhancement and refined long short-term memory," *IEEE Trans. Instrum. and Meas.*, vol. 70, pp. 1-11, Jul. 2021.

[24] D. A. Mansour, "Development of a new graphical technique for dissolved gas analysis in power transformers based on the five combustible gases," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 22, no. 5, pp. 2507-2512, Oct. 2015.

[25] Q. Zhang, H. Wang, J. Dong, G. Zhong and X. Sun, "Prediction of sea surface temperature using long short-term memory," *IEEE Geosci. Remote. Sens. Lett*., vol. 14, no. 10, pp. 1745-1749, Oct. 2017.

[26] T. Li, M. Hua and X. Wu, "A hybrid CNN-LSTM model for forecasting particulate matter (PM2.5)," *IEEE Access*, vol. 8, pp. 26933-26940, Feb. 2020.

[27] S. R. Gunn, "Support vector machines for classification and regression," ISIS technical report, 1998.

[28] M. Czajkowski and M. Kretowski, "The role of decision tree representation in regression problems – an evolutionary perspective," *Appl. Soft Comput*., vol. 48, pp. 458-475, Jul. 2016.

[29] C. E. Rasmussen and C. K .I. Williams, "Regression," in *Gaussian Processes for Machine Learning*, the MIT Press, 2006, ch.2, pp. 7-29.

[30] W. Yan, "Toward automatic time-series forecasting using neural networks," *IEEE Trans. Neural Netw. Learn. Syst*., vol. 23, no. 7, pp. 1028-1039, Jul. 2012.

[31] L. Wang, T. Littler and X. Liu, "Artificial intelligence model for transformer fault diagnosis using a constructed database," in *ACDC*, Xi'an, China, 2020, pp. 1474-1479.